Finkraft B2B E-Invoicing Platform: Master CTO Submission

UAE Federal Tax Authority (FTA) Peppol 5-Corner DCTCE Compliance Platform
Candidate: Srinivas Senniappan | Submission Deliverable: Design Note

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Part 1: Architecture Design Note

1. Business Context & Explicit Assumptions

Under the UAE Federal Tax Authority (FTA) mandate, e-invoicing is governed by a Decentralized Continuous Transaction Control and Exchange (DCTCE) 5-Corner Model: C1 (Seller / Finkraft Customer) -> C2 (Seller ASP / Finkraft Integration Point) -> C3 (Buyer ASP) -> C4 (Buyer), with concurrent C5 (FTA Clearance). Invoice records represent sovereign tax documents: zero invoices may be lost, and zero duplicates may be filed.

> **Explicit Architectural Assumptions**
> * Business Key Uniqueness: Compound key (seller_trn, invoice_number) is globally unique and immutable.
> * Pre-Submission Enrichment: Finkraft validates and enriches raw customer feeds into compliant PINT AE UBL 2.1 XML prior to ASP dispatch. Schema errors (BT-48, bad TRNs) are trapped internally.
> * ASP A Profile: Synchronous POST /invoices (200), polling GET, 409 duplicate check, 10 req/s limit; network timeouts occasionally occur AFTER acceptance.
> * ASP B Profile: Asynchronous POST /v2/submissions (202), webhooks only, ZERO duplicate checks, out-of-order/duplicate webhooks.
> * ASP Switching: In-flight invoices finish tracking on originating ASP adapter; new submissions route to updated preferred ASP.

2. Unified Adapter Contract & Dynamic ASP Switching

To shield core business logic from ASP idiosyncrasies, Finkraft implements an extensible Adapter Pattern coordinated via ASPRegistry. Customers switch or add ASPs dynamically without codebase changes.

Method	Input Signature	Output Return	Operational Purpose
submit_invoice	CanonicalInvoice (PINT AE)	SubmissionResult	Dispatches payload; normalizes sync (ASP A) vs async job (ASP B).
poll_status	external_id: str	StatusUpdate	Polls status for GET ASPs (A); raises NotSupported for webhook-only ASPs.
parse_webhook	headers: dict, payload: dict	StatusUpdate	Verifies HMAC signature, deduplicates event ID, normalizes C3/C5 status.
get_capabilities	None	ASPCapabilities	Exposes rate limits (10 rps), sync/async traits, and native idempotency flags.

3. Invoice State Machine & Normalization Matrix

Finkraft manages an internal state machine that decouples internal lifecycle management from ASP-specific statuses. Milestones c3_delivered_at and c5_reported_at accumulate independently to resolve into COMPLIANT_AND_DELIVERED.

ASP A Status	ASP B Status	Normalized Status	Finkraft Internal State & Target
RECEIVED	QUEUED / ACCEPTED	ACKNOWLEDGED	ACKNOWLEDGED_C2 (Payload accepted by C2 ASP)
VALIDATED	N/A (Implicit)	VALIDATED	VALIDATED_C2 (ASP schema & PEPPOL signature

			verified)
DELIVERED (C3)	BUYER_RECEIVED (C3)	DELIVERED_C3	DELIVERED_C3 (or COMPLIANT_AND_DELIVERED if C5 already recorded)
REPORTED (C5)	TAX_CONFIRMED (C5)	REPORTED_C5	REPORTED_C5 (or COMPLIANT_AND_DELIVERED if C3 already recorded)
REJECTED	FAILED	REJECTED	REJECTED (Terminal failure; captures BT-xx schema/tax errors)

4. Failure Handling & Reliability Engineering

Failure Mode	Root Cause & Business Risk	Finkraft Technical Mitigation
Submission Timeout (ASP A)	ASP A times out AFTER invoice accepted. Naive retries duplicate invoices or fail.	Quarantine to SUBMISSION_AMBIGUOUS. Automatic retries strictly banned. Background active reconciliation queries ASP A by invoice number. If found, advances to ACKNOWLEDGED_C2; if absent, safe retry.
Duplicate Submissions (ASP B)	ASP B has ZERO duplicate checks. Re-submitting creates duplicate tax records at FTA.	Client-Side Pessimistic Idempotency Lock: DB unique index on (seller_trn, invoice_number) + atomic compare-and-swap (VALIDATED -> SUBMITTING). In-memory repository catches DuplicateInvoiceError.
Out-of-Order Webhooks (ASP B)	TAX_CONFIRMED (C5) arrives before BUYER_RECEIVED (C3).	Milestone Lattice Resolution: Tracks independent c3_delivered_at and c5_reported_at timestamps. Invoice transitions to COMPLIANT_AND_DELIVERED as soon as both milestones are satisfied.
Late / Duplicate Webhooks	ASP B re-delivers callbacks hours later.	Idempotent Webhook Event Ledger: Unique constraint on (asp_id, event_id). Duplicates return HTTP 200 immediately. Monotonic transition guards reject retroactive state regressions as safe no-ops.
ASP A Rate Limiting (10 req/s)	Bulk upload bursts exceed ASP A quota (429).	Token-Bucket Egress Throttler: Dispatcher meters traffic to 10 req/s per tenant. Excess jobs remain buffered in persistent queues with exponential jittered backoff.

5. Architectural Trade-offs & Strategic Decisions

Key Architectural Trade-offs

Decision 1: Asynchronous Unified Event-Driven Pipeline vs. Synchronous Passthrough

* Choice: Unified async ingestion for both ASPs. Bulk uploads receive immediate HTTP 202 with internal receipt.
* What We Gained: Vendor portability (customers switch ASPs without API changes); absorbs ASP A 10 req/s quota; fault isolation.
* What We Sacrificed: Single-invoice POS clients cannot get synchronous C2 clearance within the same HTTP connection.

Decision 2: Pessimistic Idempotency & Reconciliation Quarantine vs. Optimistic Fast Retries

* Choice: Strict unique index on (seller_trn, invoice_number) + quarantine state (SUBMISSION_AMBIGUOUS).
* What We Gained: 100% guarantee against duplicate tax reporting to UAE FTA (zero customer tax penalties); tamper-proof audit trail.
* What We Sacrificed: Recovery latency (ambiguous network glitches resolve via 30s recon worker rather than 1s instant retry).